ARCHITECTUUR- & PROMPTGIDS: GOOGLE ANTIGRAVITY CLI

Project: Gedragsgestuurde Gezondheidsapp voor de Amerikaanse Markt (HTML5/SPA)

Methodologie: Atomic Habits (James Clear) & Richtlijnen Voedingscentrum / Schijf van Vijf (Aangepast naar US context)

> **Status & Doel:** Dit document bevat het volledige operationele stappenplan voor de `google-antigravity` CLI interface en de bijbehorende, maximaal uitgebreide systeem-prompt om een complete, performante, en privacy-vriendelijke HTML5 applicatie te genereren. Dit document dient als directe blauwdruk voor software engineering via LLM-compilatie.

Deel 1: Strategische Context & Lokalisatie (US vs. EU)

Om een gezondheidsapp succesvol te introduceren op de Amerikaanse markt, moet de applicatie rekening houden met fundamentele sociaal-geografische en industriële verschillen ten opzichte van Europa:

- **Voedselkwaliteit en Regelgeving:** In de VS is het aandeel ultra-geproceerd voedsel (UPF) significant hoger. Ingrediënten zoals 'high-fructose corn syrup' (HFCS) en specifieke additieven (die in de EU verboden of strikt gereguleerd zijn) zijn de norm. De app moet hierop anticiperen door een strengere scoring toe te kennen aan dergelijke ingrediënten.
- **Metrieken en Eenheden:** Amerikanen gebruiken uitsluitend het imperiale stelsel. Calorieën, ounces (oz), pounds (lbs), cups, en miles/steps zijn verplicht. Een conversiemodel op de achtergrond is cruciaal voor synchronisatie met internationale standaarden.
- **Voedingsrichtlijnen:** Hoewel de Nederlandse 'Schijf van Vijf' als wetenschappelijke onderbouwing dient voor het optimale dieet (groenten, fruit, volkoren, peulvruchten, minder vlees), moet de interface dit vertalen naar herkenbare Amerikaanse equivalenten (bijv. MyPlate-structuur), om vervreemding van de doelgroep te voorkomen.

Deel 2: Het Volledige Stappenplan voor Google Antigravity CLI

Volg exact deze stappen binnen de terminalomgeving om de applicatie te initialiseren, configureren, genereren en valideren.

Stap 1: Omgevingsvoorbereiding & CLI Installatie (5 minuten)

Zorg dat de nieuwste versie van de Google Antigravity toolchain is geïnstalleerd. Open de terminal en voer uit:

```
npm install -g @google/antigravity-cli
antigravity auth:login
antigravity project:init us-health-app --template=vanilla-spa
```

Stap 2: Requirements & Randvoorwaarden Laden (5 minuten)

Maak een configuratiebestand genaamd `antigravity.config.json` in de rootmap om de harde prestatiecriteria vast te leggen:

```
{
  "target": "html5-pwa",
  "optimization": {
    "maxBundleSizeKB": 5000,
    "uiTransitionLimitMS": 200,
    "budgetDeviceFriendly": true
  },
  "compatibility": ["iOS-WebView", "Android-WebView"],
  "license": "MIT-Open-Source"
}
```

Stap 3: De Mega-Prompt Uitvoeren (10 minuten)

Gebruik de compilatie-engine van Antigravity om de volledige applicatie-structuur in één keer te genereren op basis van de gedetailleerde prompt uit Deel 3:

```
antigravity generate:app --prompt-file=mega_prompt.txt --output=./dist
```

Stap 4: Lokale Validatie & Performantietests (5 minuten)

Controleer of de applicatie voldoet aan de harde criteria (< 2 seconden interface-wisseling en correcte offline werking via LocalStorage):

```
antigravity test:performance ./dist
antigravity serve ./dist --open
```

Deel 3: De Maximaal Uitgebreide Systeem-Prompt

Kopieer de onderstaande tekst integraal naar het bestand `mega_prompt.txt` *om het te voeren aan de Google Antigravity CLI.*

```
================================================================================
SYSTEM PROMPT FOR GOOGLE ANTIGRAVITY CLI: COMPREHENSIVE BEHAVIORAL HEALTH SPA
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ROLE & TARGET ARCHITECTURE:
You are an expert Frontend Architect and Behavioral Psychologist specializing in the
"Atomic Habits" framework by James Clear. Generate a single, production-ready, fully
self-contained HTML5 Single Page Application (SPA).
- Technology stack: Pure Vanilla HTML5, CSS3 (with responsive modern variables), and
Vanilla JavaScript (ES6+). No external framework dependencies (No React, No Angular, No
Vue).
```

```
- Compatibility: Cross-platform responsive layout optimized for iOS WebView and Android
WebView containers.
- Performance: Maximum bundle size equivalent under 500MB (keep asset footprints
minimal). UI view switching must execute in under 200ms. Code must run fluidly on low-end
budget smartphones.
- Privacy & Trust: Open-source architecture. Store 100% of user behavior, steps, and
caloric data locally in the browser using encrypted or structured LocalStorage. No third-
party tracking or server-side mandates.

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CORE FRAMEWORK IMPLEMENTATION: ATOMIC HABITS METHODOLOGY
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1. PILLAR 1: MAKE IT OBVIOUS (MAAK HET OVERDUIDELIJK)
   - Baseline Confrontation Feature: Upon first launch, the app must lock into a 1-day
evaluation mode. The user logs their current standard diet. The app assigns a nutrition
quality 'Score' (based on the Dutch "Schijf van Vijf" adapted for US items) to clearly
expose ultra-processed vulnerabilities.
   - Daily Schedule Alignment: An onboarding wizard asks the user for their typical daily
meal times (Breakfast, Lunch, Dinner, Snacks). The app uses these timestamps to schedule
local web push notification triggers.
   - Habit Stacking Engine: A specialized configuration screen where users input an
existing anchor habit (e.g., "Waking up", "Brushing teeth") and chain a new healthy habit
directly to it. Example UI text output: "After [Anchor Habit], I will immediately
[Healthy Habit]".
   - Environment Priming Module: A visual card deck providing explicit actionable
environmental hacks tailored to the US context (e.g., "Move high-fructose snacks to the
basement locker; place fresh fruit bowl directly on the kitchen counter in eyesight",
"Prepare tomorrow's gym clothes right next to your bed").

2. PILLAR 2: MAKE IT ATTRACTIVE (MAAK HET AANTREKKELIJK)
   - Temptation Bundling Framework: A feature allowing users to pair an action they
*need* to do (e.g., prepping a healthy lunch) with an action they *want* to do (e.g.,
listening to a specific entertainment podcast or watching a show).
   - Micro-Culture Leaderboard Simulation: Since the app is offline-first, generate an
interactive local leaderboard with simulated peers (e.g., "Sarah from Austin", "John from
Chicago") to construct a social norming effect around clean eating.
   - Motivation Ritual Trigger: A checklist item that enforces a 2-minute relaxation or
high-pleasure activity immediately *before* executing a difficult dietary habit.

3. PILLAR 3: MAKE IT EASY (MAAK HET GEMAKKELIJK)
   - Friction Reduction Wizard: Provides 1-click meal-prep blueprints and automated
dynamic US grocery shopping lists based on whole foods, reducing steps to healthy
execution.
   - The 2-Minute Rule Starter: An interface element that shrinks massive lifestyle goals
into 2-minute launch versions (e.g., Instead of "Do a 60-minute workout", it locks in
"Put on running shoes and walk for 2 minutes").

4. PILLAR 4: MAKE IT SATISFYING (MAAK HET BEVREDIGEND)
   - Immediate Visual Reward System: Upon clicking "Complete Meal" or "Hit Step Goal",
trigger a localized CSS confetti animation combined with an immediate point accumulation
feedback loop.
   - Advanced Streak & Calendar Tracker: A calendar matrix displaying a visual streak
```

```
counter. Healthy days are highlighted in emerald green.
   - "Never Miss Twice" Protection Logic: If the local database detects a user failed to
hit their target yesterday, the application sends a high-priority alert today with
modified, easier parameters to explicitly prevent back-to-back failures.

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FUNCTIONAL SECTIONS & UI COMPONENT SPECIFICATIONS
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The single-page structure must include an immutable, elegant bottom navigation bar
switching instantly between these views:

A. VIEW 1: DASHBOARD (THE PERFECT DIET SKETCH)
   - A comprehensive multi-nutrient progress wheel showing target macro-nutrients based
on US Dietary norms (Carbohydrates, Proteins, Fats, Fiber, Sodium tracking).
   - Live differential displays showing exactly how many units the user is away from
their optimal whole-food intake.
   - A contextual recipe-push card recommending immediate, easy meals to balance out
current nutrient deficits.

B. VIEW 2: LOGGING & CALORIE/STEP COUNTER
   - Dual-input mechanisms: A step-counter logger (simulating hardware pedometer sync via
LocalStorage incrementor) and a localized US calorie dictionary search bar (using
Imperial ounces, cups, and item packs).

C. VIEW 3: HABIT ARCHITECT (ATOMIC TRACKER)
   - Interface for habit-stacking configuration, active streaks visualization, and
environment priming tips toggle.

D. VIEW 4: OPEN-SOURCE TRANSPARENCY & PRIVACY
   - A dedicated tab exposing the raw JSON data payload currently saved in LocalStorage,
with an explicit statement explaining data ownership and open-source validation.

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TECHNICAL SPECIFICATIONS & CSS DESIGN SYSTEM
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- Design Language: Modern, clean, minimalist. Palette: Deep Navy (#1b4f72) for primary
hierarchy, Muted Emerald (#27ae60) for success/health metrics, Soft Light Slate (#f4f6f7)
for backgrounds, and Charcoal (#2c3e50) for readable text.
- Text Sizes: Headers 16-18pt, Subheaders 12-14pt, Body text 10.5pt.
- Code Requirements: Provide all semantic CSS styles inside a `
```